

# TLQLDMR: A Transfer Learning Quantile Regression Model for Wind Power Forecasting under Extreme Weather Conditions

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## ARTICLE INFO

### Keywords:

Wind power forecasting  
Large margin distribution machine  
Quantile Regression  
Transfer learning

## ABSTRACT

Accurate wind power forecasting under extreme weather conditions is pivotal for grid security but remains a formidable challenge due to the **scarcity of extreme samples** and **violent distribution shifts** (drifts). Conventional data-driven models, constrained by the independent and identically distributed (i.i.d.) assumption, often fail to generalize when facing these sporadic “Black Swan” events. Furthermore, existing transfer learning approaches largely focus on unsupervised feature alignment, neglecting the critical diagnostic value of scarce target labels, while standard large margin regression methods enforce symmetric error distributions that fundamentally contradict the asymmetric requirements of probabilistic interval forecasting.

To bridge these gaps, this paper proposes a novel **Transfer Learning Quantile Large Margin Distribution Machine (TLQLDMR)**. First, we theoretically resolve the conflict between robust learning and quantile property by constructing an **Asymmetric Variance Regularizer**. This innovation compacts the error distribution to enhance robustness while preserving the necessary skewness for accurate quantile estimation. Second, we establish a **Supervised Transfer Learning** framework that integrates dual-space adaptation. Beyond aligning feature distributions via Maximum Mean Discrepancy (MMD) in the input space, we introduce a **Weighted Pinball Loss** in the output space. This mechanism treats scarce extreme weather samples as high-weight “Corrective Anchors”, forcing the model to rectify prediction biases caused by dominant normal patterns. Comprehensive experiments demonstrate that TLQLDMR significantly outperforms state-of-the-art baselines, generating tighter and more reliable prediction intervals that effectively quantify uncertainty in extreme meteorological scenarios.

## 1. Introduction

## 2. Related Work

This section briefly reviews the theoretical foundations related to the proposed method, including Support Vector Quantile Regression, Large Margin Distribution Machine for Regression, and Transfer Learning with Maximum Mean Discrepancy.

### 2.1. Support Vector Quantile Regression (SVQR)

Traditional Support Vector Regression (SVR) relies on the  $\epsilon$ -insensitive loss function, which assumes a symmetric error distribution (e.g., Gaussian) and estimates the conditional mean of the target variable. However, in scenarios like partial shading or wind curtailment, the noise distribution is often asymmetric and heteroscedastic. To address this, Support Vector Quantile Regression (SVQR) was introduced to estimate conditional quantiles, providing a comprehensive view of the outcome distribution.

The core of SVQR is the non-symmetric pinball loss function, defined as:

$$L_{\tau}(u) = \begin{cases} \tau u, & u \geq 0 \\ (\tau - 1)u, & u < 0, \end{cases} \quad (1)$$

where  $\tau \in (0, 1)$  indicates the target quantile. Unlike SVR, SVQR finds a decision boundary that divides the data such that a proportion  $\tau$  of the samples lie below the boundary. The optimization problem minimizes the structural risk

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along with the empirical risk measured by the pinball loss:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n L_{\tau}(y_i - (\mathbf{w}^T \phi(\mathbf{x}_i) + b)), \quad (2)$$

this formulation allows SVQR to capture heavy-tailed distributions and outliers more effectively than mean-based regression.

## 2.2. Large Margin Distribution Machine for Regression (LDMR)

While SVR and SVQR focus on maximizing the minimum margin (the distance from the closest instances to the hyperplane), they largely ignore the statistical distribution of the margin over the entire dataset. The **Large Margin Distribution Machine (LDMR)** was proposed to improve generalization by explicitly optimizing the distribution of the margins (or residuals).

The key idea of LDMR is to not only maximize the margin of the support vectors but also to minimize the variance of the error distribution for all samples. A typical LDMR formulation for regression integrates two additional statistical regularization terms into the objective function:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + \lambda_1 \text{Var}(\text{Residuals}) + \lambda_2 \text{Mean}(\text{Residuals}) + C \sum_{i=1}^n L(y_i, f(\mathbf{x}_i)), \quad (3)$$

by constraining the variance, LDMR forces the prediction errors to be clustered tightly, leading to a more compact and robust hypothesis space. Our work extends this concept by modifying the symmetric variance term of standard LDMR to an asymmetric one suitable for quantile regression.

## 2.3. Transfer Learning and Maximum Mean Discrepancy (MMD)

Standard machine learning models operate under the assumption that training (source) and testing (target) data are drawn from the same probability distribution. When this assumption is violated—such as in wind power forecasting across different weather regimes—the model performance degrades significantly. Transfer learning aims to transfer knowledge from a label-rich source domain to a target domain with different but related distributions.

A fundamental challenge in transfer learning is measuring and reducing the distribution discrepancy. Maximum Mean Discrepancy (MMD) is a widely used non-parametric metric that measures the distance between two distributions in a Reproducing Kernel Hilbert Space (RKHS). Given source data  $\{\mathbf{x}_{S,i}\}$  and target data  $\{\mathbf{x}_{T,j}\}$ , the squared MMD distance is defined as:

$$\text{MMD}^2(D_S, D_T) = \left\| \frac{1}{n_S} \sum_{i=1}^{n_S} \phi(\mathbf{x}_{S,i}) - \frac{1}{n_T} \sum_{j=1}^{n_T} \phi(\mathbf{x}_{T,j}) \right\|_{\mathcal{H}}^2, \quad (4)$$

where  $\phi(\cdot)$  is the feature mapping induced by the kernel. By minimizing the MMD distance during training, the model learns a feature representation where the source and target domains are statistically aligned, enabling the classifier or regressor trained on the source domain to generalize well to the target domain.

## 3. Proposed Methodology

In this section, we elaborate on the construction of the Transfer Learning Quantile Regression Model (TLQLDMR). We first define the necessary notations, then present the final objective function which integrates Structural Risk, Distribution Variance, Domain Adaptation (MMD), and Weighted Pinball Loss. Finally, we describe the constraints introduced to handle the non-differentiability of the Quantile Loss.

### 3.1. Notations

In the field of machine learning, a domain  $\mathcal{D}$  consists of a feature space  $\mathcal{X}$  and a marginal probabilistic distribution  $P(\mathbf{X})$  on the feature space, i.e.,  $\mathcal{D} = \{\mathcal{X}, P(\mathbf{X})\}$ . Given a domain, a task  $\mathcal{T}$  consists of a label space  $\mathcal{Y}$  and an objective prediction function  $f(\cdot)$ , represented as  $\mathcal{T} = \{\mathcal{Y}, f(\cdot)\}$ . From the perspective of probability theory, the objective prediction function can be expressed as  $P(Y|\mathbf{X})$ , and the task is represented as  $\mathcal{T} = \{\mathcal{Y}, P(Y|\mathbf{X})\}$ .

In the problem of wind power forecasting under extreme weather, there are two crucial concepts corresponding to the different data distributions. The **Source Domain (Normal Weather)** is represented as  $D_S = \{(\mathbf{x}_{S,i}, y_{S,i})\}_{i=1}^{n_S}$ ,

containing a large amount of conventional weather samples. The **Target Domain (Extreme Weather)** is represented as  $D_T = \{(\mathbf{x}_{T,j}, y_{T,j})\}_{j=1}^{n_T}$ , containing a small number of extreme weather samples. Here, the input space  $\mathcal{X} \subseteq \mathbb{R}^d$  represents the feature vectors constructed after sliding window processing, and the output space  $\mathcal{Y} \subseteq \mathbb{R}$  represents the wind power.  $n_S$  and  $n_T$  denote the number of samples in the source and target domains, respectively, satisfying  $n_T \ll n_S$ .

Although the source and target domains belong to the same type of forecasting task, their data distributions exhibit distinct characteristics due to the drastic meteorological changes during extreme events, i.e.,  $P_S(\mathbf{X}_S, \mathbf{Y}_S) \neq P_T(\mathbf{X}_T, \mathbf{Y}_T)$ . This distribution discrepancy implies that a model trained solely on  $P_S$  will fail on  $P_T$ . We define a feature mapping  $\phi : \mathcal{X} \rightarrow \mathcal{H}$  that maps the input to a high-dimensional Reproducing Kernel Hilbert Space (RKHS). The prediction model is defined as:

$$f(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x}) + b, \quad (5)$$

where  $\mathbf{w}$  is the weight vector and  $b$  is the bias. The quantile  $\tau \in (0, 1)$  is set to a specific value (e.g.,  $\tau = 0.95$ ) for interval prediction (e.g., predicting the upper bound).

### 3.2. TLQLDMR Model Formulation

#### 3.2.1. Final Objective Function

Our optimization goal is to find the optimal  $\mathbf{w}, b$  and slack variables  $\xi, \xi^*$  to minimize the following objective function  $J$ :

$$\begin{aligned} \min_{\mathbf{w}, b, \xi, \xi^*} J = & \underbrace{\frac{1}{2} \|\mathbf{w}\|^2}_{(1) \text{ Structural Risk}} \\ & + \underbrace{\lambda_1 \sum_{i=1}^{n_S} (y_{S,i} - \mathbf{w}^T \phi(\mathbf{x}_{S,i}) - b - \bar{u}_S)^2}_{(2) \text{ Distribution Variance (LDMR Core)}} \\ & + \underbrace{\lambda_2 \left\| \frac{1}{n_S} \sum_{i=1}^{n_S} \phi(\mathbf{x}_{S,i}) - \frac{1}{n_T} \sum_{j=1}^{n_T} \phi(\mathbf{x}_{T,j}) \right\|_{\mathcal{H}}^2}_{(3) \text{ Domain Adaptation (MMD)}} \\ & + \underbrace{C_S \sum_{i=1}^{n_S} L_{\tau}(y_{S,i}, f(\mathbf{x}_{S,i})) + C_T \sum_{j=1}^{n_T} L_{\tau}(y_{T,j}, f(\mathbf{x}_{T,j}))}_{(4) \text{ Weighted Pinball Loss}}, \end{aligned} \quad (6)$$

the physical meaning of each term is as follows:

1. **Structural Risk:** This term controls the complexity of the model and prevents overfitting, embodying the maximum margin principle of SVM.
2. **Distribution Variance:** This term is the core of our proposed Quantile LDMR (QLDMR), incorporating a critical modification to the original LDMR variance.
  - **Theoretical Conflict:** The original LDMR minimizes the variance of absolute residuals, which assumes a symmetric error distribution suitable for mean regression. However, this symmetry contradicts the inherent asymmetry of Quantile Regression (governed by  $\tau$ ). Applying a symmetric variance penalty would force the residual distribution towards the center, degrading the skewness required for accurate quantile estimation.
  - **Asymmetric Modification:** To resolve this, we employ an **Asymmetric Variance** regularization based on the algebraic (signed) residuals. This design allows the model to reduce error dispersion (compactness) without constraining the distribution's direction, thereby preserving the non-symmetric properties essential for the Pinball Loss and improving interval coverage.
3. **Domain Adaptation (MMD):** In the context of wind power forecasting under extreme weather conditions, our primary objective is to mitigate the differences in data distributions across various domains by means of learning a feature transformation. Consequently, the effective measurement of distribution disparities between

domains becomes a paramount concern. In this pursuit, it is crucial to select a method suitable for quantifying distribution dissimilarity, while ensuring that the chosen method is computationally efficient and easily integrated into the optimization problem of our model. Maximum Mean Discrepancy (MMD) has been demonstrated as an effective approach for estimating the distance between two distributions, offering a relatively straightforward implementation with fast computation speed. It targets the input space  $\mathcal{X}$  to minimize the distance between the distribution centers of the source and target domains.

4. **Weighted Pinball Loss:** This term targets the output space  $\mathcal{Y}$  and acts as the decisive component for “Supervised Transfer Learning”. Unlike unsupervised transfer which relies solely on feature alignment, we explicitly incorporate the target domain samples  $D_T$  into the loss calculation for the following reasons:

- **Corrective Anchors:** The target samples act as “anchors for correction”. The source domain data ( $D_S$ ) teaches the model general laws (e.g., higher wind speed leads to higher power), but extreme weather samples ( $D_T$ ) often violate these laws (e.g., high wind speed causing cut-out and zero power). If  $D_T$  were excluded from the loss, the model, driven implicitly by  $D_S$ , would wrongly predict full power during extreme events. Including  $D_T$  in the loss imposes a heavy penalty for such mispredictions, forcing the model to learn the specific mapping of extreme events.
- **Differentiation from MMD:** While MMD brings the “appearance” (features) of normal and extreme weather closer, the Pinball Loss ensures the “content” (labels) is accurate. It forces the regression curve to correctly envelope the rare data points that MMD cannot correct on its own.
- **Asymmetric Weighted Strategy:** We employ a weighted strategy to address the extreme sample imbalance ( $n_T \ll n_S$ ). By setting  $C_T \gg C_S$  (e.g.,  $C_T = 100C_S$ ), we essentially tell the model that “mispredicting one extreme sample is as grave as mispredicting a hundred normal samples”. This forces the optimizer to prioritize the scarce extreme data, preventing them from being drowned out by the massive conventional data.

### 3.2.2. Constraints

To handle the non-differentiability of the Pinball Loss, we introduce slack variables  $\xi_k, \xi_k^*$  to transform it into linear constraints. For every sample  $k$  in the training set  $D = D_S \cup D_T$ :

$$\begin{cases} y_k - (\mathbf{w}^T \phi(\mathbf{x}_k) + b) \leq \xi_k, & \forall k \in \{1, \dots, n_S + n_T\} \\ (\mathbf{w}^T \phi(\mathbf{x}_k) + b) - y_k \leq \xi_k^*, & \forall k \in \{1, \dots, n_S + n_T\} \\ \xi_k \geq 0, \quad \xi_k^* \geq 0, \end{cases} \quad (7)$$

in the objective function, the term  $L_\tau(y_k, f(\mathbf{x}_k))$  corresponds to  $\tau \xi_k + (1 - \tau) \xi_k^*$ .

## 3.3. Model Optimization

In this section, we derive the dual problem of TLQLDMR to enable efficient solving using Quadratic Programming (QP).

### 3.3.1. Matrix Reformulation of the Primal Problem

To facilitate the derivation, we rewrite the variance term and the MMD term in the objective function [Eq. (6)] using matrix traces or norms. Let  $X = [X_S, X_T]$  denote the entire training set with  $N = n_S + n_T$  samples. For consistency with standard derivations, we add a coefficient of  $1/2$  to the variance and MMD terms. The primal objective can be reformulated as:

$$\min_{\mathbf{w}, b, \xi, \xi^*} \quad \frac{1}{2} \|\mathbf{w}\|^2 + \frac{\lambda_1}{2} \text{Var}_S(\mathbf{w}, b) + \frac{\lambda_2}{2} \text{MMD}^2(\mathbf{w}) + \sum_{k=1}^N C_k L_\tau(y_k, f(\mathbf{x}_k)) \quad (8)$$

where  $C_k$  takes value  $C_S$  if sample  $k \in D_S$  and  $C_T$  if  $k \in D_T$ .

We introduce two auxiliary matrices to express the variance and MMD terms in Eq. (8) as quadratic forms of the weights:

1. **Variance Matrix  $\mathbf{S}$ :** To represent the source domain variance  $\sum_{i \in D_S} (y_{S,i} - f(\mathbf{x}_{S,i}) - \bar{u}_S)^2$ , we define a centering matrix  $\mathbf{H}_S = \mathbf{I}_{n_S} - \frac{1}{n_S} \mathbf{1}_{n_S} \mathbf{1}_{n_S}^T$ . The variance term can be written as  $\|\mathbf{H}_S(\mathbf{y}_S - \mathbf{f}_S)\|^2$ . To extend this to the full

dataset, we define an  $N \times N$  matrix  $\mathbf{S}$ :

$$\mathbf{S} = \begin{bmatrix} \mathbf{H}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \quad (9)$$

2. **MMD Matrix  $\mathbf{M}$ :** The MMD term represents the distance between means. We define the MMD coefficient matrix  $\mathbf{M} \in \mathbb{R}^{N \times N}$  as:

$$M_{ij} = \begin{cases} \frac{1}{n_S^2} & i, j \in D_S \\ \frac{1}{n_T^2} & i, j \in D_T \\ \frac{-1}{n_S n_T} & \text{otherwise} \end{cases} \quad (10)$$

### 3.3.2. Representer Theorem and Kernelization

According to the Representer Theorem [Belkin et al. \(2006\)](#), the optimal weight vector  $\mathbf{w}$  lies in the span of the feature mappings of the training samples:

$$\mathbf{w} = \sum_{i=1}^N \beta_i \phi(\mathbf{x}_i) = \Phi \boldsymbol{\beta} \quad (11)$$

where  $\Phi = [\phi(\mathbf{x}_1), \dots, \phi(\mathbf{x}_N)]$  is the feature matrix and  $\boldsymbol{\beta} = [\beta_1, \dots, \beta_N]^T$  are the coefficients. Substituting Eq. (11) into the prediction function, we get  $f(\mathbf{X}) = \boldsymbol{\beta}^T \mathbf{K} + b \mathbf{1}^T$ , where  $\mathbf{K} = \Phi^T \Phi$  is the  $N \times N$  kernel matrix. Consequently, using Eq. (9) and Eq. (10), the structural risk, variance, and MMD terms can be expressed in terms of  $\boldsymbol{\beta}$ :

- Structural Risk:  $\|\mathbf{w}\|^2 = \boldsymbol{\beta}^T \mathbf{K} \boldsymbol{\beta}$ .
- Distribution Variance: The quadratic part with respect to  $\boldsymbol{\beta}$  is  $\boldsymbol{\beta}^T \mathbf{K} \mathbf{S} \mathbf{K} \boldsymbol{\beta}$ . Note that the bias  $b$  is eliminated by the centering matrix  $\mathbf{H}_S$ .
- Domain Adaptation (MMD):  $\text{MMD}^2(\mathbf{w}) = \boldsymbol{\beta}^T \mathbf{K} \mathbf{M} \mathbf{K} \boldsymbol{\beta}$ .

Combining these, we define the total coefficient matrix  $\boldsymbol{\Omega}$ :

$$\boldsymbol{\Omega} = \mathbf{K} + \lambda_1 \mathbf{K} \mathbf{S} \mathbf{K} + \lambda_2 \mathbf{K} \mathbf{M} \mathbf{K} \quad (12)$$

By substituting the kernelized terms and Eq. (12) into Eq. (8), the kernelized primal problem becomes (ignoring constant terms w.r.t  $b$  in variance):

$$\min_{\boldsymbol{\beta}, b, \xi, \xi^*} \quad \frac{1}{2} \boldsymbol{\beta}^T \boldsymbol{\Omega} \boldsymbol{\beta} - (\lambda_1 \mathbf{K} \mathbf{S} \mathbf{Y})^T \boldsymbol{\beta} + \sum_{k=1}^N C_k(\tau \xi_k + (1 - \tau) \xi_k^*) \quad (13)$$

subject to the constraints from Eq. (7):

$$y_k - (\mathbf{K}_k \boldsymbol{\beta} + b) \leq \xi_k, \quad (\mathbf{K}_k \boldsymbol{\beta} + b) - y_k \leq \xi_k^*, \quad \xi_k, \xi_k^* \geq 0 \quad (14)$$

### 3.3.3. Lagrangian and Dual Problem

We verify the problem by introducing Lagrange multipliers  $\alpha_k, \alpha_k^*, \mu_k, \mu_k^* \geq 0$  for Eq. (14). The Lagrangian function  $\mathcal{L}$  is constructed as:

$$\begin{aligned} \mathcal{L} = & \frac{1}{2} \boldsymbol{\beta}^T \boldsymbol{\Omega} \boldsymbol{\beta} - (\lambda_1 \mathbf{K} \mathbf{S} \mathbf{Y})^T \boldsymbol{\beta} + \sum_{k=1}^N C_k(\tau \xi_k + (1 - \tau) \xi_k^*) \\ & - \sum_{k=1}^N \alpha_k (\xi_k - y_k + \mathbf{K}_k \boldsymbol{\beta} + b) - \sum_{k=1}^N \alpha_k^* (\xi_k^* + y_k - \mathbf{K}_k \boldsymbol{\beta} - b) - \sum (\mu_k \xi_k + \mu_k^* \xi_k^*) \end{aligned} \quad (15)$$

By setting the partial derivatives of Eq. (15) with respect to the primal variables  $(b, \xi, \xi^*, \boldsymbol{\beta})$  to zero, we obtain the KKT conditions:

$$\frac{\partial \mathcal{L}}{\partial b} = 0 \implies \sum_{k=1}^N (\alpha_k - \alpha_k^*) = 0 \quad (16)$$

$$\frac{\partial \mathcal{L}}{\partial \xi_k} = 0, \frac{\partial \mathcal{L}}{\partial \xi_k^*} = 0 \implies \alpha_k \in [0, C_k \tau], \quad \alpha_k^* \in [0, C_k(1 - \tau)] \quad (17)$$

$$\frac{\partial \mathcal{L}}{\partial \beta} = 0 \implies \Omega \beta - \lambda_1 KSY - K(\alpha - \alpha^*) = 0 \quad (18)$$

Let  $\gamma = \alpha - \alpha^*$ . From Eq. (18), the optimal  $\beta$  is derived as:

$$\beta = \Omega^{-1}(K\gamma + \lambda_1 KSY) \quad (19)$$

Substituting Eq. (19) back into Eq. (15), and simplifying using Eq. (16) and Eq. (17), we arrive at the **Dual Problem**, which is a standard Quadratic Programming (QP) problem:

$$\begin{aligned} \min_{\alpha, \alpha^*} \quad & \frac{1}{2}(\alpha - \alpha^*)^T \mathbf{Q}_{dual}(\alpha - \alpha^*) + \mathbf{p}_{dual}^T(\alpha - \alpha^*) \\ \text{s.t.} \quad & \sum_{k=1}^N (\alpha_k - \alpha_k^*) = 0 \\ & 0 \leq \alpha_k \leq C_k \tau, \quad \forall k \\ & 0 \leq \alpha_k^* \leq C_k(1 - \tau), \quad \forall k \end{aligned} \quad (20)$$

where the dual matrix  $\mathbf{Q}_{dual}$  and vector  $\mathbf{p}_{dual}$  are defined as:

$$\mathbf{Q}_{dual} = K\Omega^{-1}K, \quad \mathbf{p}_{dual} = \lambda_1 K\Omega^{-1}KSY - Y \quad (21)$$

Once the optimal  $\alpha, \alpha^*$  are found using a QP solver for Eq. (20), we can recover  $\beta$  using Eq. (19) and construct the final decision function  $f(\mathbf{x})$ .

### 3.4. The Algorithm and Complexity Analysis

The complete training process of the proposed TLQLDMR model is summarized in Algorithm 1. The algorithm mainly consists of matrix construction and solving a Quadratic Programming (QP) problem.

#### 3.4.1. Complexity Analysis

The computational complexity of TLQLDMR primarily stems from three parts: matrix multiplication, matrix inversion, and the QP solver.

1. **Matrix Operations:** The construction of  $\mathbf{K}$ ,  $\mathbf{S}$ , and  $\mathbf{M}$  involves  $O(N^2d)$  or  $O(N^2)$  operations. The calculation of  $\Omega$  involves matrix multiplications of size  $N \times N$ , leading to a complexity of  $O(N^3)$ .
2. **Matrix Inversion:** Inverting the  $N \times N$  matrix  $\Omega$  requires  $O(N^3)$  complexity.
3. **QP Solver:** Solving the standard Quadratic Programming problem for the dual variables typically has a complexity ranging from  $O(N^{2.3})$  to  $O(N^3)$  depending on the solver implementation (e.g., Interior Point Method).

Therefore, the overall time complexity of the proposed algorithm is approximately  $O(N^3)$ . Given that  $N = n_S + n_T$  and  $n_T \ll n_S$ , the scale is dominated by the source domain sample size. For large-scale datasets, approximation techniques like the Nyström method can be employed to reduce the complexity, which will be considered in future work.

## 4. Experiments

## 5. Conclusion

### CRedit authorship contribution statement

\*\*\*: Methodology, Software, Writing - original draft. \*\*\*: Visualization, Data curation. \*\*\*: Formal analysis, Software. \*\*\*: Writing - review & editing. \*\*\*: Funding acquisition, Validation, Supervision.

### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

**Algorithm 1:** TLQLDMR Training Algorithm

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**Input** : Source Data  $D_S = \{(\mathbf{x}_{S,i}, y_{S,i})\}_{i=1}^{n_S}$ , Target Data  $D_T = \{(\mathbf{x}_{T,j}, y_{T,j})\}_{j=1}^{n_T}$ , Quantile  $\tau$ , Kernel parameters, Regularization parameters  $\lambda_1, \lambda_2$ , Pinball Loss weights  $C_S, C_T$ .

**Output:** Optimal dual variables  $\alpha, \alpha^*$  and bias  $b$ .

// Step 1: Data Preparation

- 1 Combine Source and Target data:  $\mathbf{X} = [\mathbf{X}_S, \mathbf{X}_T]$ ,  $\mathbf{Y} = [\mathbf{Y}_S; \mathbf{Y}_T]$ ,  $N = n_S + n_T$ ;
- 2 Set weights vector  $\mathbf{C} \in \mathbb{R}^N$  where  $C_k = C_S$  for  $k \leq n_S$  and  $C_T$  otherwise;

// Step 2: Matrix Construction

- 3 Compute Kernel Matrix  $\mathbf{K} \in \mathbb{R}^{N \times N}$  using the specified kernel function;
- 4 Construct Variance Matrix  $\mathbf{S}$  using Eq. (9);
- 5 Construct MMD Matrix  $\mathbf{M}$  using Eq. (10);

// Step 3: Dual Problem Formulation

- 6 Compute coefficient matrix  $\mathbf{\Omega} = \mathbf{K} + \lambda_1 \mathbf{K} \mathbf{S} \mathbf{K} + \lambda_2 \mathbf{K} \mathbf{M} \mathbf{K}$  using Eq. (12);
- 7 Invert  $\mathbf{\Omega}$  to fit the dual form;
- 8 Compute Dual Matrix  $\mathbf{Q}_{dual} = \mathbf{K} \mathbf{\Omega}^{-1} \mathbf{K}$  using Eq. (21);
- 9 Compute Dual Vector  $\mathbf{p}_{dual} = \lambda_1 \mathbf{K} \mathbf{\Omega}^{-1} \mathbf{K} \mathbf{S} \mathbf{Y} - \mathbf{Y}$  using Eq. (21);

// Step 4: Solve QP

- 10 Solve the QP problem in Eq. (20) providing  $\mathbf{Q}_{dual}$ ,  $\mathbf{p}_{dual}$ , and constraints  $0 \leq \alpha_k \leq C_k \tau$ ,  $0 \leq \alpha_k^* \leq C_k(1 - \tau)$ ;
- 11 **Return** Optimal  $\alpha, \alpha^*$ ;

// Step 5: Recover Primal Variables

- 12 Compute  $\gamma = \alpha - \alpha^*$ ;
- 13 Recover  $\beta$  using Eq. (19);
- 14 Compute bias  $b$  using complementary slackness conditions (averaged over Support Vectors);

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**Acknowledgments**

This work is supported by the National Key Research and Development Program of China (No. 2021YFB3101500) and the Chongqing Municipal Training Program of Innovation and Entrepreneurship for Undergraduate (No. X202310635204).

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